The empirical recurrence for OEIS A236030, recovered from its tail

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Abstract

OEIS A236030 records "Empirical recurrence of order 92" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1848$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 96$. Its last coefficient is $c_{92} = 905969664 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A236030 is "Number of (n+1) X (3+1) 0..2 arrays colored with the difference of the upper median and the lower median in each 2 X 2 subblock.". It has offset 1 and begins

$$1848, 14980, 116786, 965344, 8289258, 68613596, 570818506, 4747089488, \ldots$$

The entry states:

> Empirical recurrence of order 92 (see link above).

As of the "Last modified" line on the live entry (Apr 25 2023 14:43:12, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1848$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1848$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3696$ exact terms from $n = 4$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{92} c_i\, a(n-i),$$

c_{1}	0	c_{24}	−44148928558257	c_{47}	−973481789662598406	c_{70}	−583979889006623616
c_{2}	40	c_{25}	−5316709348720	c_{48}	−20836031452336768	c_{71}	−5314186079381955392
c_{3}	0	c_{26}	7657840132474	c_{49}	2170804032639492016	c_{72}	4713903624829871184
c_{4}	2373	c_{27}	103835197188918	c_{50}	−29344263985119264	c_{73}	369167055566253504
c_{5}	446	c_{28}	887166998557535	c_{51}	1638644176727755406	c_{74}	−1584702852243534272
c_{6}	−9815	c_{29}	30525069301628	c_{52}	604769325235116038	c_{75}	805070630970901120
c_{7}	−294	c_{30}	506647873726715	c_{53}	−4327513395416007956	c_{76}	−313186959218024448
c_{8}	−1153326	c_{31}	−1761315383945000	c_{54}	−191664002759730033	c_{77}	38112526852068608
c_{9}	−303298	c_{32}	−9985097339519955	c_{55}	−623479269534479546	c_{78}	102303561655459328
c_{10}	1691008	c_{33}	377965136651276	c_{56}	−3284217564430039506	c_{79}	−134549174611658752
c_{11}	2198860	c_{34}	−9892015404583778	c_{57}	5350839016291381314	c_{80}	41029874608390656
c_{12}	229277836	c_{35}	15968958859640122	c_{58}	−752784201911095652	c_{81}	−8054873416687616
c_{13}	59598992	c_{36}	60420821793635484	c_{59}	−1656295368308703400	c_{82}	15602595543040
c_{14}	−304501551	c_{37}	−153993489429304 02	c_{60}	2829321596900728180	c_{83}	5419166823735296
c_{15}	−617593334	c_{38}	57391291499858102	c_{61}	2287786863894318000	c_{84}	−1848032969809920
c_{16}	−23097112888	c_{39}	−83321726712674406	c_{62}	−1070555226014961979	c_{85}	1301071370747904
c_{17}	−5295653278	c_{40}	−207691695923631809	c_{63}	1190273551782062648	c_{86}	371860839956480
c_{18}	29910499812	c_{41}	180462893401742490	c_{64}	4963396996972191988	c_{87}	230595255074816
c_{19}	67083194908	c_{42}	−133449316352005587	c_{65}	−1105875454396408460 0	c_{88}	100544516456448
c_{20}	1309031637784	c_{43}	281313239190594640	c_{66}	2437041837727260820	c_{89}	13722177241088
c_{21}	244609737416	c_{44}	309154203817102175	c_{67}	3703265321822678080	c_{90}	1658562019328
c_{22}	−1211884968862	c_{45}	−781606184250628244	c_{68}	−8102335337441796964	c_{91}	−509758930944
c_{23}	−3541052156032	c_{46}	171980599734939562	c_{69}	6510722007234587888	c_{92}	905969664

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 96$, and it is the recurrence OEIS A236030 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1 x^{92-1} - \cdots - c_{92}$. Its constant term is $-c_{92} = -905969664$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 905969664.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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